

Comparison for W3 and 13

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```
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(tidyr)
library(purrr)
library(stringr)
library(ggplot2)
library(clue)

m_w3_nocov <- readRDS("lca_w3_k4_nocov.rds")
m_w13_nocov <- readRDS("lca_w13_k4_nocov_update.rds")
m_w3_cov <- readRDS("lca_w3_k4_cov.rds")
m_w13_cov <- readRDS("lca_w13_k4_cov_update.rds")

# Fit comparison (AIC/BIC/G2, N, K)
fit_row <- function(m, wave, cov){
  tibble(
    wave = wave,
    covars = cov,
    k = ncol(m$P),
    N = m$N,
    AIC = m$aic,
    BIC = m$bic,
    G2 = m$Gsqr
  )
}

fit_cmp <- bind_rows(
  fit_row(m_w3_nocov, "W3", "no"),
  fit_row(m_w13_nocov, "W13", "no"),
  fit_row(m_w3_cov, "W3", "yes"),
  fit_row(m_w13_cov, "W13", "yes")
)

fit_cmp

## # A tibble: 4 x 6
```

```
##   wave  covars      N    AIC    BIC    G2
##   <chr> <chr> <int> <dbl> <dbl> <dbl>
## 1 W3    no      16351 61044. 61314. 289.
## 2 W13   no      13700 53943. 54206. 313.
## 3 W3    yes     16351 58226. 58911. 686.
## 4 W13   yes     13700 51149. 51819. 583.

# Class size comparison (estimated mixing proportions)
mix_row <- function(m, wave, cov){
  tibble(wave = wave, covars = cov, class = seq_along(m$P), proportion = as.numeric(m$P))
}
mix_cmp <- bind_rows(
  mix_row(m_w3_nocov, "W3", "no"),
  mix_row(m_w13_nocov, "W13", "no"),
  mix_row(m_w3_cov, "W3", "yes"),
  mix_row(m_w13_cov, "W13", "yes")
)
mix_cmp

## # A tibble: 16 x 4
##   wave  covars class proportion
##   <chr> <chr> <int>     <dbl>
## 1 W3    no      1      0.0147
## 2 W3    no      2      0.797
## 3 W3    no      3      0.149
## 4 W3    no      4      0.0385
## 5 W13   no      1      0.166
## 6 W13   no      2      0.0183
## 7 W13   no      3      0.777
## 8 W13   no      4      0.0379
## 9 W3    yes     1      0.0239
## 10 W3   yes     2      0.757
## 11 W3   yes     3      0.0275
## 12 W3   yes     4      0.192
## 13 W13   yes     1      0.270
## 14 W13   yes     2      0.0481
## 15 W13   yes     3      0.661
## 16 W13   yes     4      0.0208

# Quick entropy / classification sharpness
entropy <- function(p) -sum(p * log(p + 1e-12))
norm_entropy <- function(post) {
  # average normalized entropy across cases (0 = sharp, 1 = diffuse)
  K <- ncol(post); Hmax <- log(K)
  mean(apply(post, 1, entropy) / Hmax)
}
sharpness <- function(post) mean(apply(post, 1, max)) # closer to 1 is better

qual_cmp <- tibble(
  wave = c("W3", "W13", "W3", "W13"),
  covars = c("no", "no", "yes", "yes"),
  norm_entropy = c(norm_entropy(m_w3_nocov$posterior),
    norm_entropy(m_w13_nocov$posterior),
    norm_entropy(m_w3_cov$posterior),
    norm_entropy(m_w13_cov$posterior)),

```

```

max_p_mean = c(sharpness(m_w3_nocov$posterior),
                sharpness(m_w13_nocov$posterior),
                sharpness(m_w3_cov$posterior),
                sharpness(m_w13_cov$posterior))
)
qual_cmp

```

```

## # A tibble: 4 x 4
##   wave covars norm_entropy max_p_mean
##   <chr> <chr>         <dbl>     <dbl>
## 1 W3    no             0.215     0.891
## 2 W13   no             0.245     0.867
## 3 W3    yes            0.196     0.896
## 4 W13   yes            0.212     0.881

```

We evaluated classification quality using normalized entropy (lower = better) and the mean maximum posterior probability (higher = better).

For Wave 3, the no-covariate model showed clearer class separation than the covariate model (entropy: 0.215 vs 0.308; max posterior: 0.891 vs 0.851). This suggests that covariates introduced additional variability and slightly diluted classification certainty. In contrast, for Wave 13, adding covariates improved classification quality (entropy: 0.245 to 0.207; max posterior: 0.867 to 0.886), indicating that class membership in Wave 13 is more strongly structured by the covariates.

Do the proportions of respondents in each class differ between 1996 and 2016?

```

# match classes
flatten_probs <- function(m){
  mats <- lapply(m$probs, function(x) as.matrix(x))
  P <- do.call(cbind, mats)
  rownames(P) <- paste0("c", seq_len(nrow(P)))
  P
}
P_w3 <- flatten_probs(m_w3_nocov)
P_w13 <- flatten_probs(m_w13_nocov)

# Cosine similarity function
cos_sim <- function(A, B){
  A <- as.matrix(A); B <- as.matrix(B)
  num <- A %*% t(B)
  A_n <- sqrt(rowSums(A^2)); B_n <- sqrt(rowSums(B^2))
  sweep(sweep(num, 1, A_n, "/"), 2, B_n, "/")
}

S <- cos_sim(P_w3, P_w13) # [K x K] similarity matrix

# Optional: clean up numerical weirdness
S[is.na(S)] <- 0

# 2) Convert similarity → nonnegative cost
cost <- max(S) - S

# 3) Hungarian algorithm on cost
assign <- clue::solve_LSAP(cost)

matched_pairs <- tibble(

```

```

w3_class = seq_len(nrow(S)),
w13_class = as.integer(assign),
similarity = S[cbind(seq_len(nrow(S)), as.integer(assign))]
)

```

```
matched_pairs
```

```

## # A tibble: 4 x 3
##   w3_class w13_class similarity
##   <int>    <int>    <dbl>
## 1       1       2    0.986
## 2       2       3    0.999
## 3       3       1    0.993
## 4       4       4    0.971

```

To compare latent classes across waves, we aligned Wave 3 and Wave 13 classes based on cosine similarity of their item-response probability profiles.

High similarity indicates that the classes represent very similar underlying subpopulations.

```

# W3 proportions (no cov)
p_w3 <- tibble(
  w3_class = seq_along(m_w3_nocov$P),
  prop_w3 = as.numeric(m_w3_nocov$P)
)

# W13 proportions (no cov)
p_w13 <- tibble(
  w13_class = seq_along(m_w13_nocov$P),
  prop_w13 = as.numeric(m_w13_nocov$P)
)

# Merge using your matched_pairs table
prop_compare <- matched_pairs %>%
  left_join(p_w3, by = "w3_class") %>%
  left_join(p_w13, by = "w13_class")

prop_compare

```

```

## # A tibble: 4 x 5
##   w3_class w13_class similarity prop_w3 prop_w13
##   <int>    <int>    <dbl>    <dbl>    <dbl>
## 1       1       2    0.986    0.0147    0.0183
## 2       2       3    0.999    0.797    0.777
## 3       3       1    0.993    0.149    0.166
## 4       4       4    0.971    0.0385    0.0379

```

Which demographic covariates predict class membership, and do these associations differ across years?

```

coef_w3 <- m_w3_cov$coeff
coef_w13 <- m_w13_cov$coeff
coef_w3

```

```

##               [,1]      [,2]      [,3]
## (Intercept)  0.9815236110 -0.7236228364  1.1985445824

```

```
## age -0.0014315012 0.0005313769 -0.0007589214
## factor(gender)2 0.8077256954 0.7970673262 1.2167073328
## factor(degree)1 0.9787203159 -0.4441534759 -0.5331295437
## factor(degree)2 2.9173487920 0.9220569097 1.0985276858
## factor(degree)3 2.8629663144 -2.5271436058 -0.3530719937
## factor(degree)4 1.3584654802 -0.3498166864 -2.3267509033
## factor(degree)5 2.6332695825 -2.4420742573 -0.3131063297
## factor(degree)6 2.2838056937 -2.2507122354 0.0268857669
## factor(degree)9 0.0057578004 -0.0108508164 0.0106959353
## factor(race)1 0.9959389532 -0.5812313459 -0.1509554774
## factor(race)2 0.0712908861 0.6904060912 1.3638514984
## factor(race)7 -0.0933196248 -0.8059918239 -0.0163351193
## factor(usborn)5 -1.9938112084 -0.9499049047 -0.9311669447
## factor(usborn)9 -0.0003085191 -0.0463364801 0.0343398354
## factor(hispanic)1 -1.7434272576 -0.8366692916 -1.4098535657
## factor(hispanic)2 0.3026604724 -1.4496418050 0.8686982135
## factor(hispanic)3 0.1735255648 -0.1178445205 0.0132015477
## factor(hispanic)5 2.2435780898 1.7201285090 1.7184404053
```

coef_w13

```
##          [,1]      [,2]      [,3]
## (Intercept) -10.64105331 -6.75778263 3.96828014
## age          0.12845490 0.01317688 -0.18335298
## factor(gender)2 -0.43957080 -0.32189245 3.37310491
## factor(degree)1 0.07002859 1.12917089 -0.03915765
## factor(degree)2 0.15999215 1.85934504 -3.84353709
## factor(degree)3 0.88914555 2.70580882 12.18694832
## factor(degree)4 1.40872395 3.71276216 2.91335281
## factor(degree)5 1.84696161 4.12755432 3.60086156
## factor(degree)6 1.39433263 3.91239227 5.52621947
## factor(degree)9 1.04378630 2.80263171 -3.00038410
## factor(race)1 1.70612364 1.98580092 -3.59062659
## factor(race)2 1.20649503 0.55067870 -5.65048559
## factor(race)7 1.54693823 0.93100277 -13.54308284
## factor(usborn)5 -0.95264829 -1.68815023 -9.90995576
## factor(usborn)9 -3.73785896 -2.82841007 1.73911277
## factor(hispanic)1 -1.78348242 2.95285075 -8.13449262
## factor(hispanic)2 -2.12236794 3.04284335 1.72416285
## factor(hispanic)3 -9.50667168 4.33321691 4.50546503
## factor(hispanic)5 -1.90136186 4.06512414 -0.67129086
```